

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO5 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)</i>		
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Section / Batch:	CSE AI	Date:	

Code of Conduct

I G chandra sekhar reddy(192472173) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: g chandra sekhar reddy

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Frequency Domain Image Enhancement	Periodic noise removal, Fourier Transform, notch filtering, quality inspection	1	10
Spatial Filtering & Edge-Preserving Enhancement	Bilateral filtering, edge preservation, autonomous vehicle navigation	2	10
GRAND TOTAL:		20 Marks	

Annexure A – Scenario Based Assessment

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

- (a) Interpret the scenario and identify the type of noise present.
- (b) Explain how this degradation affects inspection accuracy.
- (c) Recommend an appropriate enhancement technique to remove the interference.
- (d) Justify why your selected approach is more suitable than spatial-domain methods.
- (e) Explain your answer in a logical sequence.

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

- (a) Interpret the scenario and explain the challenge involved.
- (b) Identify the importance of preserving edges while removing noise.
- (c) Recommend a suitable filtering technique.
- (d) Justify why your selected filter preserves important image details.
- (e) Present your explanation clearly and systematically.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
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Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

COMPUTER VISION – SCENARIO BASED ASSESSMENT

Question 1: Manufacturing Quality Inspection

a) Type of noise

The repetitive horizontal stripes observed in the captured images are caused by ****periodic noise****. Periodic noise usually occurs due to electrical interference from power supplies, electromagnetic disturbances, faulty imaging sensors, or nearby industrial machinery. Unlike random noise, periodic noise follows a regular repeating pattern across the image. Since manufacturing industries use several electrically operated machines, such interference is common in automated inspection systems. The regular horizontal bands reduce the visual quality of the image and make it difficult for computer vision algorithms to correctly analyze the product surface.

b) Effect on inspection accuracy

Periodic noise significantly affects the performance of an automated quality inspection system. Its effects include:

- The horizontal stripes introduce artificial patterns that are not actually present on the metal surface.
- Small manufacturing defects such as scratches, cracks, dents, pits, corrosion marks, and surface irregularities may become hidden behind the noise.
- The image contrast decreases, making it difficult to distinguish defective regions from normal areas.
- Edge detection algorithms may incorrectly detect the noise stripes instead of actual object boundaries.
- Feature extraction becomes less reliable because the unwanted frequencies interfere with important image details.
- Machine learning or computer vision algorithms may classify noisy regions as defects, increasing false positives.

- At the same time, actual defects may be ignored because they are masked by the interference, resulting in false negatives.
- Poor inspection accuracy may lead to defective products being shipped to customers or good products being rejected unnecessarily, increasing production costs.

Therefore, removing periodic noise is an important preprocessing step before performing defect detection or quality analysis.

c) Suitable enhancement technique

The most appropriate enhancement technique is ****Fourier Transform-based frequency-domain filtering using a notch-reject filter****.

Periodic noise appears as distinct peaks in the frequency spectrum, making it easier to isolate and remove compared to spatial-domain techniques.

Steps involved:

1. Capture the original noisy image.
2. Convert the image into the frequency domain using the Discrete Fourier Transform (DFT) or Fast Fourier Transform (FFT).
3. Observe the frequency spectrum and identify the bright peaks representing the periodic horizontal interference.
4. Design and apply notch-reject filters at the identified frequency locations to suppress only the unwanted frequency components.
5. Preserve the remaining useful frequency information that contains the actual image details.
6. Apply the Inverse Fourier Transform (IDFT) to convert the filtered image back to the spatial domain.
7. Obtain an enhanced image with significantly reduced periodic noise, making defect

detection more accurate.

d) Why frequency-domain filtering is suitable

Frequency-domain filtering is the preferred solution because periodic noise has a well-defined frequency pattern. In the frequency spectrum, the unwanted stripes appear as isolated frequency components, allowing them to be removed selectively without affecting the rest of the image.

A notch filter targets only the unwanted frequencies while preserving important image information such as edges, textures, and fine surface details. In contrast, spatial-domain filters such as averaging or Gaussian filters reduce noise by smoothing the entire image. Although these filters remove some noise, they also blur edges and fine defects, making it difficult to detect tiny cracks or scratches.

Since manufacturing inspection requires high precision, preserving image details is extremely important. Frequency-domain filtering provides better accuracy because it removes only the interference while maintaining the original quality of the inspected object.

e) Logical sequence

Original noisy image

↓

Apply Fourier Transform (DFT/FFT)

↓

Convert image to frequency domain

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Identify periodic noise frequencies

↓

Apply notch-reject filter

↓

Remove unwanted frequency components

↓

Apply Inverse Fourier Transform

↓

Enhanced image with reduced noise



Accurate defect detection and quality inspection

Thus, Fourier Transform combined with notch filtering is the most suitable enhancement technique for removing periodic electrical interference in manufacturing quality inspection systems. It improves image quality while preserving important defect information, resulting in reliable automated inspection.

Question 2: Autonomous Vehicle Navigation

a) Challenge involved

An autonomous vehicle captures road images while driving in heavy rain. Rain introduces various types of image degradation, including random noise, water droplets on the camera lens, reduced visibility, reflections from wet roads, and decreased contrast. These factors make it difficult for the computer vision system to accurately detect lane markings, road boundaries, pedestrians, vehicles, traffic signs, and other obstacles.

Despite these challenging weather conditions, the autonomous driving system must continue making safe navigation decisions in real time. Therefore, the image enhancement technique should reduce noise while preserving important road features.

b) Importance of preserving edges

Edge preservation is extremely important for autonomous vehicle navigation because edges contain valuable structural information about the environment.

The importance includes:

- Lane markings guide the vehicle in staying within the correct lane.
- Road edges help identify the boundaries of the driving area.
- Traffic signs and signals contain sharp edges that are required for accurate recognition.

- Vehicles, pedestrians, cyclists, and obstacles are detected mainly through their boundaries.
- Accurate edge information improves object detection, segmentation, and tracking algorithms.
- Blurred edges may cause incorrect lane detection, increasing the possibility of navigation errors.
- Preserving edges also improves the performance of advanced computer vision tasks such as semantic segmentation, obstacle avoidance, and path planning.

Therefore, the selected filtering technique should remove noise while maintaining the sharpness of important edges.

c) Suitable filtering technique

The recommended image enhancement technique is ****Bilateral Filtering****.

Bilateral filtering is an edge-preserving smoothing filter that effectively removes noise without significantly blurring object boundaries. Unlike traditional smoothing filters, bilateral filtering considers both the spatial relationship between neighboring pixels and their intensity similarity.

This makes it highly suitable for autonomous driving applications where image clarity and edge preservation are equally important.

d) Why bilateral filtering is suitable

Bilateral filtering provides an excellent balance between noise reduction and edge preservation.

Its advantages include:

- It smooths only pixels that have similar intensity values.
- Pixels located across strong edges are treated separately, preventing edge blurring.

- It effectively reduces noise caused by rain, sensor imperfections, and low-light conditions.
- Lane markings remain clear and well defined.
- Road boundaries are preserved for accurate navigation.
- Important object edges such as pedestrians, vehicles, and traffic signs remain sharp.
- The filter improves the performance of edge detection, feature extraction, and object recognition algorithms.
- It produces visually smoother images while maintaining structural details required for real-time driving decisions.

Compared with ordinary averaging or Gaussian filters, bilateral filtering achieves better visual quality because it preserves essential image features instead of smoothing everything uniformly.

e) Logical sequence

Rainy road image



Image acquisition



Apply bilateral filtering



Reduce rain-induced noise



Preserve edges and lane markings



Improve image clarity



Detect lanes, road boundaries, pedestrians, and vehicles



Reliable navigation and safe autonomous driving

Therefore, bilateral filtering is the most suitable enhancement technique for autonomous vehicle navigation under rainy conditions. It successfully removes unwanted noise while preserving critical edge information required for lane detection, obstacle recognition, and safe vehicle control. By maintaining important image details, bilateral filtering improves the overall accuracy and reliability of computer vision systems used in self-driving vehicles.